

System Documentation

Electronic Equipment

Automation System PowerLine

V 4000 R41

V 4000 Rx3

V 4000 Rx4

12 V 1600 R50

E532828/00E



Power. Passion. Partnership.

© 2015 Copyright MTU Friedrichshafen GmbH

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of MTU Friedrichshafen GmbH. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. MTU Friedrichshafen GmbH reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1	Safety	
1.1	Important provisions for all products	5
1.2	Personnel and organizational requirements	7
1.3	Safety regulations for startup and operation	8
1.4	Safety regulations for maintenance and repair work	9
1.5	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials	12
1.6	Standards for safety notices in the text	14
2	Purpose	
2.1	PowerLine – Use	15
2.2	Automation system PowerLine – Device overview	16
3	PowerLine – Devices	
3.1	Engine Control Unit ECU	17
3.1.1	Engine Control Unit ECU – Use and design	17
3.1.2	Engine Control Unit ECU – Function	20
3.1.3	Engine Control Unit ECU – Installation location	22
3.1.4	ECU 4 – Technical data	23
3.1.5	ECU 7 – Technical data	24
3.1.6	ECU 9 – Technical data	26
3.1.7	MTU system cables for Engine Control Unit ECU – Use and design	28
3.1.8	MTU system cables for Engine Control Unit ECU – Installation location	33
3.1.9	MTU system cable W003/W004 (ECU 4) – Technical data	34
3.1.10	MTU system cable W001/W005 (ECU 7) – Technical data	35
3.1.11	MTU system cable W001/W005 (ECU 9) – Technical data	36
3.2	Power Automation Unit PAU	37
3.2.1	Power Automation Unit PAU Engine – Scope of delivery	37
3.2.2	Power Automation Unit PAU Engine – Mechanical design	38
3.2.3	Power Automation Unit PAU Engine – Electrical design	39
3.2.4	Power Automation Unit PAU Engine – Function	40
3.2.5	Power Automation Unit PAU Engine – Installation location	46
3.2.6	Power Automation Unit PAU Engine – Connection technology	47
3.2.7	Power Automation Unit PAU Engine – Technical data	48
3.3	Power Output Module (POM)	49
3.3.1	Power Output Modul POM – Design and function	49
3.3.2	Power Output Modul POM – Installation location	50
3.3.3	Power Output Modul POM – Technical data	51
3.4	Display Instruments	52
3.4.1	Display instruments – Overview	52
3.4.2	Display instruments – Installation location	54
3.4.3	Engine speed display instrument – Design	55
3.4.4	Coolant temperature display instrument – Design	56
3.4.5	Lube oil pressure display instrument – Design	57
3.4.6	Runtime meter – Design	58
3.4.7	Engine speed display instrument – Technical data	59
3.4.8	Coolant temperature display instrument – Technical data	60
3.4.9	Lube oil pressure display instrument – Technical data	61
3.4.10	Runtime meter – Technical data	62
3.5	DC/DC Converter	63
3.5.1	DC/DC converter – Design	63
3.5.2	DC/DC converter – Function	64
3.5.3	DC/DC converter – Installation location	68
3.5.4	DC/DC converter – Technical data	69
3.6	Capacitor Power System CaPoS	70
3.6.1	Capacitor Power System CaPoS – Design	70
3.6.2	Capacitor Power System CaPoS – Function	72
3.6.3	UltraCap modules – Installation location	73
3.6.4	CaPoS modules – Technical data	74
3.6.5	UltraCap modules – Scope of delivery	75
3.6.6	Discharger – Design	76
4	Diagnostic Software Automation Suite	
4.1	Diagnostic software Automation Suite – Scope of delivery and use	77
5	Installation Specifications	
5.1	EMC regulations	78
5.2	Grounding concept	79
5.3	Insulation testing and dimensioning of lines	97
6	Functional Safety	
6.1	PowerLine – Configuration example	98
6.2	Safety features of PowerLine	99
6.3	Hazard logbook	100

7 Appendix A		7.3 MTU contact persons/service partners	110
7.1 Abbreviations	102	8 Appendix B	
7.2 Conversion tables	106	8.1 Index	111

1 Safety

1.1 Important provisions for all products

Nameplate

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. When operating the machine in the EU: The second nameplate must be affixed in a prominent position as described in the accompanying specifications.

General information

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the safety instructions and warning notices

Correct use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine control/parameters)
- In compliance with all safety regulations and in adherence with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use is considered improper use. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

Emission regulations and emission labels

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission control units/systems may only be maintained, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these regulations will invalidate the design type approval issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations. The maintenance schedules of the manufacturer must be observed over the entire life cycle of the product.

Replacing components with emission labels

Emission labels are attached to all MTU engines. These must remain on the engine throughout its operational life.

Engines used exclusively in land-based, military applications other than by US government agencies are excepted from this proviso.

Please note the following when replacing components with emission labels:

- Appropriate emission labels must be affixed on spare parts.
- Emission labels may not be transferred from old components to new ones.
- Emission labels on old components must be removed and destroyed.

1.2 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Personnel must receive instruction on product handling and maintenance on the basis of this manual with a special emphasis on safety requirements and warnings.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. safety footwear, ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety regulations for startup and operation

Safety requirements for startup

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before putting the product into service. All necessary approvals must be granted by the relevant authorities and all requirements for initial startup must be fulfilled.

When putting the product into operation, always ensure that

- All personnel is clear of the danger zone surrounding moving parts of the machine.
Electrically-actuated linkages may be set in motion when the Engine Control Unit is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All safety equipment is in place.
- No persons wearing pacemakers or any other technical body aids are present.
- The service room is adequately ventilated.
- Keep clear of the service room during the first operating hours. Hazardous gases may occur as a result of the combustion of paints or oils.
- The exhaust system is leak-tight and that the gases are vented to atmosphere.
- Protect battery terminals, generator terminals or cables against accidental contact.
- Check that all connections have been correctly allocated (e.g. +/- polarity, direction).

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be permanently observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

The procedures for emergencies, in particular, emergency stop, must be practiced regularly.

The following steps must be taken if a malfunction of the system is detected or reported by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not remain in the operating room when the product is running for any longer than absolutely necessary.

Keep a safe distance away from the product if possible. Do not touch the product unless expressly instructed to do so following a written procedure.

Do not inhale the exhaust gases of the product.

The following requirements must be fulfilled before the product is started:

- Wear ear protectors.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binder agent.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Observe the safety instructions for these devices.

1.4 Safety regulations for maintenance and repair work

Safety regulations prior to maintenance and repair work

Have maintenance or repair work carried out by qualified and authorized personnel only.

Allow the product to cool down to less than 50 °C (risk of explosion for oil vapors, fluids and lubricants, risk of burning).

Relieve pressure in fluid and lubricant systems and compressed-air lines which are to be opened. Use suitable collecting vessels of adequate capacity to catch fluids and lubricants.

When changing the oil or working on the fuel system, ensure that the service room is adequately ventilated.

Never carry out maintenance and repair work with the product in operation, unless:

- It is expressly permitted to do so following a written procedure.
- The product is running in the low load range and only for as long as absolutely necessary.

Lock-out the product to preclude undesired starting, e.g.

- Start interlock
- With hydraulic starting system: shut off supply line.

Attach "Do not operate" sign in the operating area or to control equipment.

Disconnect the battery. Lock out circuit breakers.

Close the main valve on the compressed-air system and vent the compressed-air line when air starters are fitted.

Disconnect the control equipment from the product.

Use special tools if they are specified for the relevant work.

Elastomer components (e.g. engine mounts, damping elements, couplings and V-belts) must not be painted. They may only be installed after painting the engine or must be covered before painting work is carried out.

The following applies to starters with copper-beryllium alloy pinions:

- Wear a respirator mask (filter class P3). Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the flywheel housing inside with a class H dust extraction device.
- Observe the safety data sheet.

Safety regulations during maintenance and repair work

Take special care when removing ventilation or plug screws from the product. Cover the screw or plug with a rag to prevent fluids escaping under pressure.

Take care when draining hot fluids and lubricants (risk of burning).

Use only proper and calibrated tools. Observe the specified tightening torques during assembly or disassembly.

Carry out work only on assemblies or plants which are properly secured.

Make sure components or assemblies are placed on stable surfaces. Adopt suitable measures to avoid that components/tools fall down. Use the specified lifting equipment for all components.

Never use lines for climbing.

Keep fuel injection lines and connections clean.

Carry out appropriate cleaning procedures to clean and inspect components requiring special cleanness (e.g. components carrying oil, fuel, or air).

Always seal connections with caps or covers if a line is removed or opened.

Fit new seals when re-installing lines.

Avoid damaging lines, particularly the fuel lines.

Ensure that all retainers and dampers are installed correctly.

Ensure that O-rings are not installed in a slanted/twisted condition.

Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Do not place fuel or oil lines near hot components.

Do not touch elastomeric seals (e.g. Viton sealing rings) with your bare hands if they have a carbonized or resinous appearance.

Note cooling time for components which are heated for installation or removal (risk of burning).

When working high on the engine, always use suitable ladders and work platforms.

Pay particular attention to cleanliness at all times.

Remove any condensate from components which were chilled before assembly. If necessary, coat the components with a suitable corrosion inhibitor.

Safety regulations after completion of maintenance and repair work

Before barring, make sure that nobody is standing in the danger zone of the product.

Check that all access ports/apertures which have been opened to facilitate working are closed again.

Check that all safety equipment has been installed and that all tools and loose parts have been removed (especially the barring gear).

Ensure that no unattached parts have been left in/on the product (e.g. including rags and cable straps).

Welding work

Welding operations on the product or mounted units are not permitted. Cover the product when welding in its vicinity.

Before starting welding work:

- Switch off the power supply master switch.
- Disconnect the battery.
- Separate the electrical ground of electronic equipment from the ground of the unit.

No other maintenance or repair work must be carried out in the vicinity of the product while welding is going on. Risk of explosion or fire due to oil vapors and highly flammable process materials.

Do not use product as ground terminal.

Never position the welding power supply cable adjacent to, or crossing wiring harnesses of the product. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Remove components (e.g. exhaust pipe) from the product before performing necessary welding work.

Hydraulic installation and removal

Check satisfactory function and safe operating condition of tools, jigs and fixtures to be used. Use only the specified jigs and fixtures for hydraulic removal/installation procedures.

Observe the max. permissible force-on pressure specified for the jig/fixture.

Do not attempt to bend or exert force on HP lines.

Before starting work, pay attention to the following:

- Vent the installation/removal jig, the pumps and the pipework at the relevant designated points.
- During the installation procedure, screw on jig with plunger extended.
- During the removal procedure, screw on jig with plunger retracted.

For a hydraulic installation/removal jig with central expansion pressure supply, screw spindle into shaft end until correct sealing is established.

During hydraulic installation/removal of components, ensure that no persons are in the direct vicinity of the component being pressed.